

Waleska Lopez

Professor Nafa Fatema

EECE2140

Classwork 2

Pseudocode:

- 1st declare an integer for the menu
- 2nd begin a do while loop
 - Inside this loop, declare menu
- User choice
- Begin statement beased on choice
 - Declare seqSize and make realresult an empty string
 - Display, ask user how many ## in secret code
 - Input that to seqSize
 - While seqSize is ≤ 0 , show error ask for +# and input seqSize
- For I in 0 – seqSize
 - Show enter number, input int code, and if between 1 – 26
 - Calculate ltr by using `(char)('a' + (code-1))`.
 - Append ltr to result
- Else append ? result
- Break
- Help, explain instructions to user, break
- Exit, show exiting program and break

- Else show invalid option must choose
- Repeat until choice not 0
- End

Logic Explanation:

In task 2, for loops and mapping is user. Bounds were set for the for loop seen in $i < \text{seqSize}$ in order to make sure it matches the number of times the user inputs. It also uses the ASCII values where a is 65 so if a user enters 1 the formula will equal 65('a') and so on and so forth. This make sure the mapping matches the assignments rules. In task 3, the while loop is implemented to make sure a positive number is given by the user, this makes sure the program has valid data to work with. In task 4, the do while loop menu appears and will repeat until the user chooses 0. The switch statement organizes the different program behaviors effectively.

```
PROBLEMS OUTPUT TERMINAL ...  /secret_decoder - eece2140 +   ... |  
```

waleska@wales-laptop:~/eece2140\$./secret_decoder

```
--- Secret Decoder Menu ---
1: decode a message
2: help (instructions)
0: exit
selection: 1
how many numbers are in the secret sequence? 5
enter number 1: 8
enter number 2: 5
enter number 3: 12
enter number 4: 12
enter number 5: 15
the decoded message is: HELLO

--- Secret Decoder Menu ---
1: decode a message
2: help (instructions)
0: exit
selection: 
```

```
--- Secret Decoder Menu ---
1: decode a message
2: help (instructions)
0: exit
selection: 1
how many numbers are in the secret sequence? -2
error, invalid length. Enter a positive number: 
```

```
waleska@wales-laptop:~/eece2140$ ./secret_decoder

--- Secret Decoder Menu ---
1: decode a message
2: help (instructions)
0: exit
selection: 2
instruction, numbers 1-26 map to letters A-Z. anything else maps to '?'.

--- Secret Decoder Menu ---
1: decode a message
2: help (instructions)
0: exit
selection: 0
exiting program. bye!
```